BLOCKCHAIN-ASSISTED PRIVACY-PRESERVING AND SYNCHRONIZED KEY AGREEMENT FOR DTNS

ABSTRACT

With the rapid growth of cyber threats and unauthorized data access, traditional authentication mechanisms are insufficient to secure digital file-sharing systems. This project proposes a Blockchain-Enabled Secure File Sharing Framework integrated with Multi-Factor Biometric Authentication and a Digital Twin Network-Based Threat Intelligence Layer.The system implements a three-level authentication mechanism consisting of password verification, Local Binary Pattern Histogram (LBPH)-based facial recognition, and One-Time Password (OTP) validation. To ensure file integrity, SHA-256 cryptographic hashing generates a unique digital fingerprint for each uploaded file. All authentication events, file transactions, and hash records are stored in a private blockchain network, ensuring tamper-proof auditability and transparency.To enhance proactive cybersecurity, a Digital Twin Network (DTN) is integrated as a virtual behavioral replica of the system. The DTN monitors login failures, biometric mismatches, OTP errors..The integration of biometric authentication, blockchain immutability, cryptographic integrity verification, and real-time Digital Twin-based anomaly detection provides a resilient and intelligent secure file-sharing environment. The proposed framework enhances system transparency, predictive threat mitigation, and adaptive security enforcement, making it suitable for enterprise-level data protection and critical infrastructure security applications.

EXISTING SYSTEM

Current digital file-sharing systems primarily rely on centralized architectures where data is stored and managed on a single cloud server or enterprise database. Popular platforms such as Google Drive, Dropbox, and Microsoft OneDrive use standard username–password authentication with optional two-factor authentication (2FA) through email or SMS-based OTP verification. In enterprise environments, file sharing is commonly implemented using FTP/SFTP servers or Network Attached Storage (NAS) systems with role-based access control mechanisms.

Traditional systems focus mainly on data encryption during transmission using SSL/TLS protocols and basic password hashing techniques for credential protection. File activity logs and access records are stored in centralized databases, which are managed by system administrators. Although audit logs are maintained, they are not immutable and can be altered if the central server is compromised. Some advanced enterprise solutions incorporate monitoring tools to detect repeated login failures or suspicious IP addresses, but these mechanisms are largely reactive rather than predictive.

Existing systems generally lack biometric-based multi-factor authentication, immutable blockchain-backed storage of transaction records, and intelligent behavioral analysis for detecting abnormal usage patterns. File integrity verification is often limited to basic checksum validation, without decentralized or tamper-proof mechanisms. As a result, traditional file-sharing platforms remain vulnerable to brute-force attacks, phishing attempts, insider threats, and unauthorized data manipulation.

Overall, existing file-sharing systems provide basic confidentiality and access control but do not offer integrated blockchain immutability, biometric security layers, or proactive Digital Twin-based anomaly detection for advanced cyber threat prevention.

DISADVANTAGES

• Centralized Architecture: Single point of failure; if the server is compromised, all files and logs are at risk.

• Weak Authentication Mechanisms: Primarily password-based systems are vulnerable to brute-force, phishing, and credential-stuffing attacks.

• No Biometric Verification: Lack of facial recognition or advanced multi-factor biometric authentication.

• Non-Immutable Logs: Audit trails can be modified or deleted by administrators or attackers.

• Limited File Integrity Assurance: No blockchain-based tamper-proof hashing mechanism for verifying file authenticity.

• Reactive Security Model: Threat detection occurs only after suspicious activity is identified.

• No Behavioral Intelligence: Absence of Digital Twin or AI-driven anomaly detection for monitoring abnormal access patterns.

• Insider Threat Vulnerability: Administrators with high privileges may misuse or manipulate stored data.

PROPOSED SYSTEM

The proposed system introduces a Blockchain-Enabled Secure File Sharing Framework integrated with Multi-Factor Biometric Authentication and a Digital Twin Network-Based Threat Intelligence Layer. Unlike traditional centralized file-sharing platforms, the proposed architecture ensures tamper-proof data management, advanced identity verification, and proactive cyber threat detection within a unified security framework.

The system begins with a secure user registration and authentication process that implements a three-level verification mechanism. First, users authenticate using a username and password. Second, biometric verification is performed using Local Binary Pattern Histogram (LBPH)-based facial recognition to validate the user's identity. Third, a One-Time Password (OTP) is generated and sent to the registered email or mobile number for final authentication. This layered approach significantly reduces the risk of unauthorized access.

Once authenticated, users can upload and share files securely. Each uploaded file undergoes SHA-256 cryptographic hashing to generate a unique digital fingerprint, ensuring file integrity and preventing tampering. All authentication events, file uploads, downloads, and hash values are recorded in a private blockchain network. The blockchain ensures immutability, transparency, and decentralized verification of records, eliminating the risk of log manipulation or unauthorized alterations.

To enhance proactive cybersecurity, the system integrates a Digital Twin Network (DTN), which acts as a virtual behavioral replica of the file-sharing environment. The DTN continuously monitors user activities such as login failures, biometric mismatches,

incorrect OTP attempts, and abnormal file download patterns. By analyzing these behavioral metrics, the system detects potential brute-force attacks, credential misuse, and suspicious access attempts in real time. Upon detecting anomalies, the system triggers alerts and enforces adaptive security measures such as temporary account lockout or additional verification steps.

The framework is implemented using Python and Flask for backend development, OpenCV for facial recognition processing, cryptographic libraries for SHA-256 hashing, and a custom private blockchain module for secure transaction storage. The integration of blockchain immutability, biometric authentication, cryptographic integrity verification, and Digital Twin-based anomaly detection ensures a secure, intelligent, and enterprise-ready file-sharing solution.

ADVANTAGES

• Multi-Layer Authentication: Combines password, biometric facial recognition, and OTP verification for enhanced security.

• Tamper-Proof Audit Trail: Blockchain-based storage ensures immutable logging of all transactions and authentication events.

• File Integrity Assurance: SHA-256 hashing guarantees that uploaded files cannot be altered without detection.

• Proactive Threat Detection: Digital Twin Network continuously monitors and detects abnormal behavior in real time.

• Brute-Force Prevention: Automatically identifies repeated failed login attempts and enforces adaptive security controls.

• Transparency and Trust: Decentralized record management improves accountability and prevents log manipulation.

• Enterprise-Grade Security: Suitable for protecting sensitive organizational and critical infrastructure data.

• Adaptive Security Enforcement: Dynamically responds to suspicious activities with automated alerts and restrictions.

SYSTEM SPECIFICATION:

HARDWARE CONFIGURATION:

- Processor - i5
- Speed - 3 GHz
- RAM - 4 GB(min)
- Hard Disk - 500 GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

SOFTWARE CONFIGURATION:

- Operating System: Linux, Windows/7/10
- Tools: Anaconda, Jupiter, vscode
- Front End: HTML, CSS
- Server side Script: Python , AIML

CHAPTER 1. INTRODUCTION

In today's digitally interconnected world, secure file sharing has become a critical requirement for organizations, enterprises, and critical infrastructure systems. The rapid expansion of cloud computing, remote collaboration, and distributed networks has significantly increased the volume of sensitive data transmitted across digital platforms. However, this growth has also amplified cybersecurity risks, including unauthorized access, data breaches, identity theft, insider threats, and sophisticated brute-force attacks. Traditional authentication mechanisms that rely solely on username-password combinations are increasingly vulnerable to credential theft, phishing, replay attacks, and database compromises.

To address these evolving security challenges, advanced technologies such as blockchain, biometric authentication, and intelligent threat monitoring systems are being explored. Blockchain technology, originally introduced through Satoshi Nakamoto in the context of Bitcoin, provides a decentralized and immutable ledger system capable of ensuring tamper-proof data storage and transparent auditing. Its cryptographic foundation and distributed consensus mechanism make it highly suitable for secure transaction logging and integrity verification in file-sharing systems.

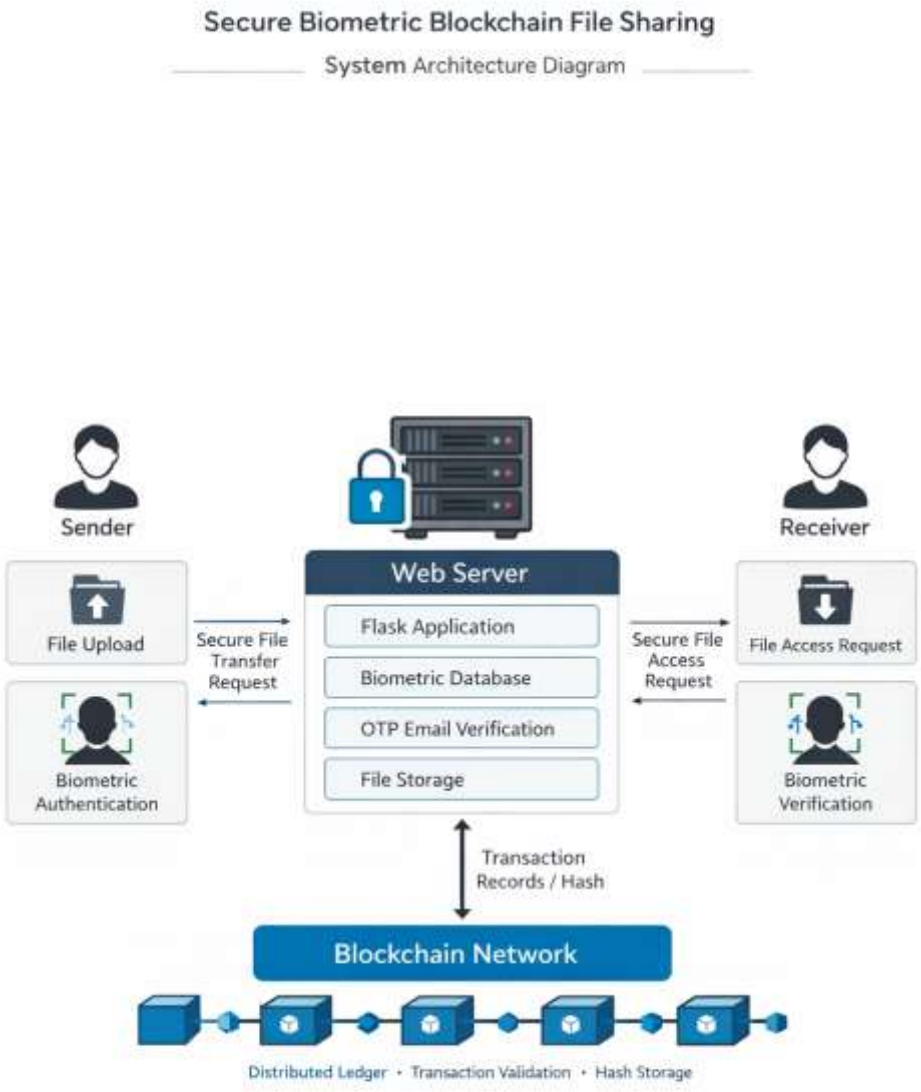
At the same time, multi-factor authentication (MFA) has emerged as a robust solution to strengthen user verification processes. By integrating password validation with biometric recognition techniques such as Local Binary Pattern Histogram (LBPH)-based facial recognition and One-Time Password (OTP) verification, systems can significantly reduce the risk of unauthorized access. Biometric authentication leverages unique physiological characteristics, providing an additional security layer that is difficult to replicate or forge.

Beyond authentication and data integrity, modern cybersecurity demands proactive threat detection. The concept of a Digital Twin Network (DTN) introduces a virtual replica of the operational system that monitors behavioral patterns, authentication attempts, and file transaction activities. By analyzing anomalies such as repeated login failures, biometric mismatches, abnormal download frequencies, or unusual access times, the system can detect potential cyber threats before they escalate into full-scale breaches.

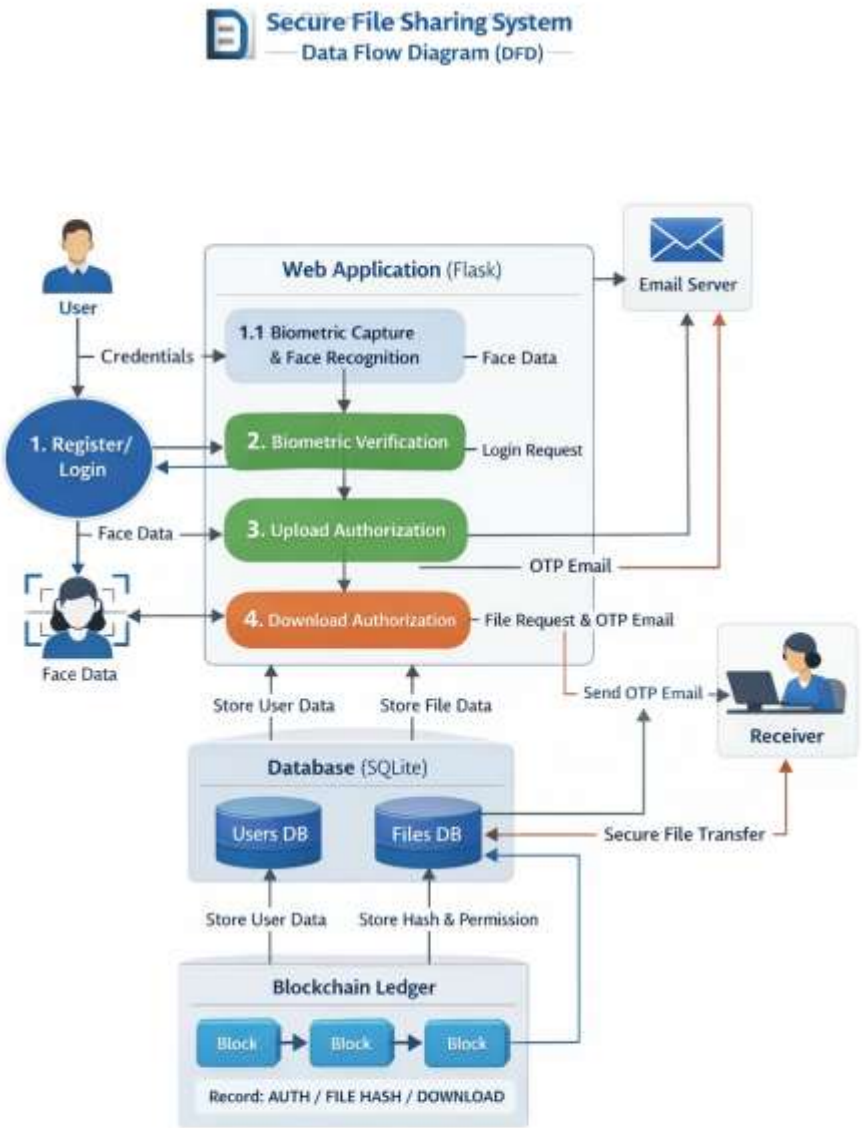
This project presents a Blockchain-Enabled Secure File Sharing Framework integrated with Multi-Factor Biometric Authentication and a Digital Twin Network-Based Threat Intelligence Layer. The framework combines cryptographic hashing (SHA-256) for file integrity, private blockchain architecture for immutable record keeping, multi-level authentication for access control, and real-time anomaly detection for proactive cybersecurity.

By integrating these advanced technologies into a unified architecture, the proposed system enhances transparency, accountability, predictive threat mitigation, and adaptive security enforcement. The framework is particularly suited for enterprise environments, government agencies, healthcare institutions, and critical infrastructure sectors where data confidentiality, integrity, and availability are of paramount importance.

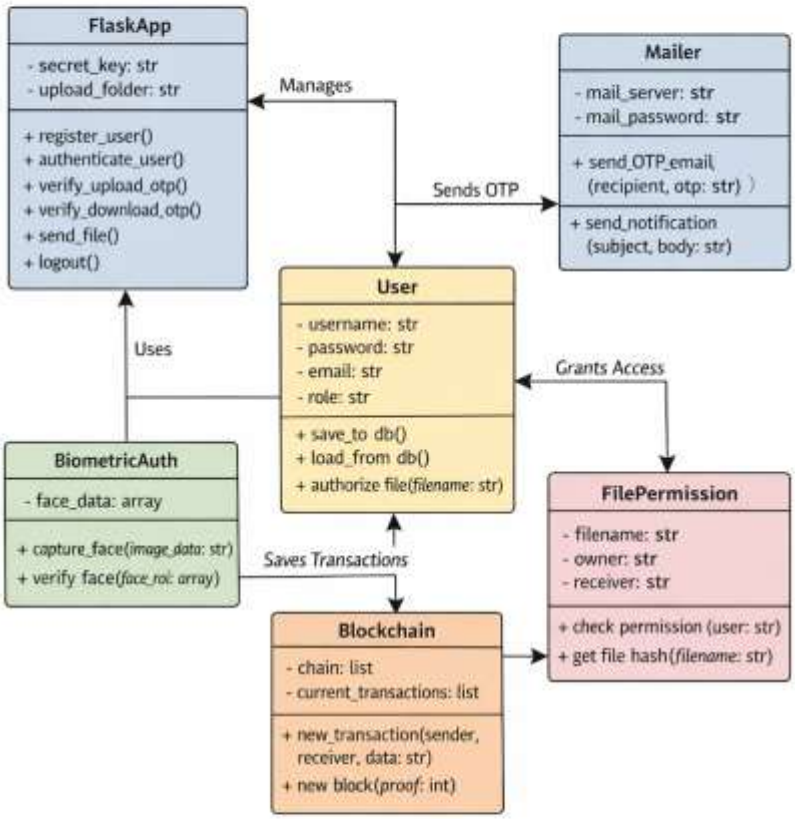
SYSTEM ARCHITECTURE DIAGRAM



DATA FLOW DIAGRAM

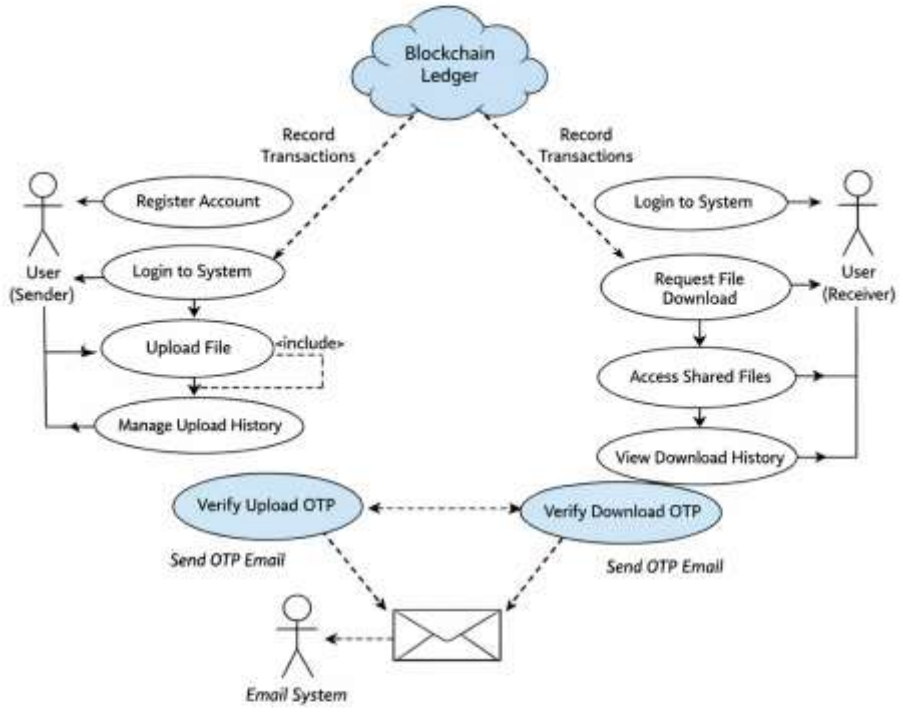


CLASS DIAGRAM

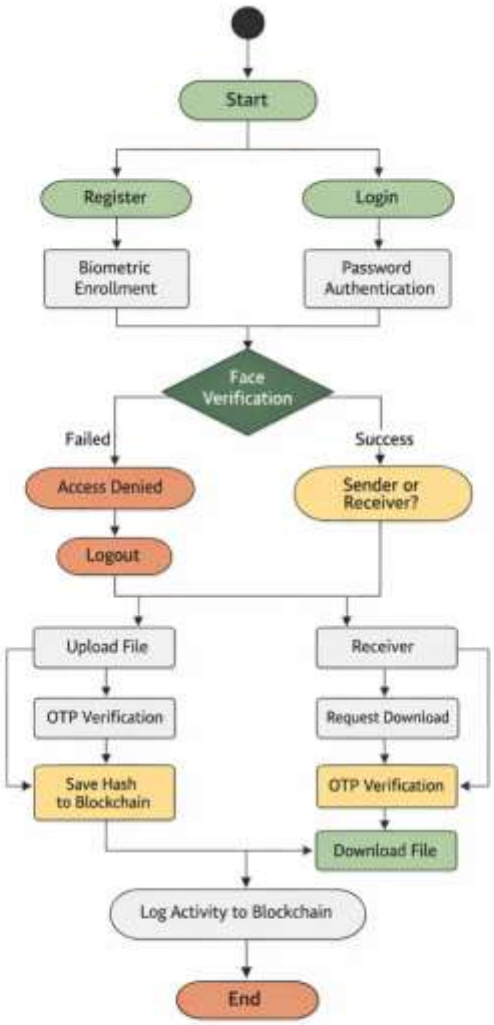


SEQUENCE DIAGRAM

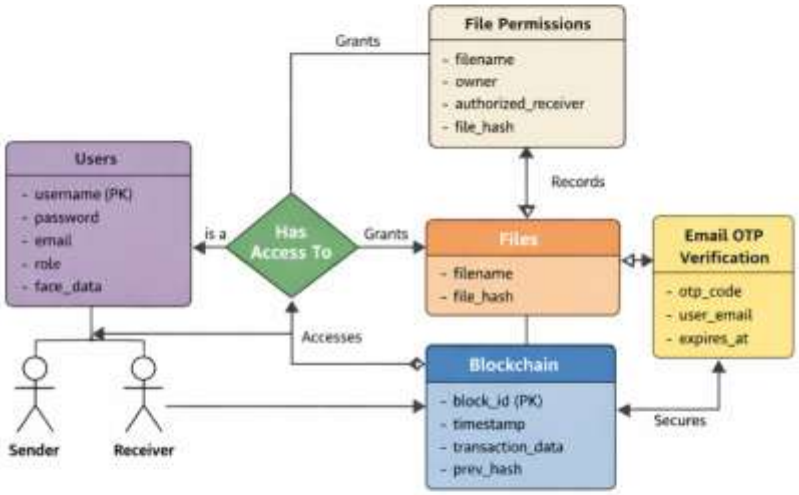
USE-CASE DIAGRAM



ACTIVITY DIAGRAM



ER-DIAGRAM


File Permissions
- filename
- owner
- authorized_receiver
- file_hash
Grants
Records
Users
- username (PK)
- password
- email
- role
- face_data
is a
Has Access To
Grants
Files
- filename
- file_hash
Email OTP Verification
- otp_code
- user_email
- expires_at
Accesses
Blockchain
- block_id (PK)
- timestamp
- transaction_data
- prev_hash
Secures
Sender
Receiver

CHAPTER 2. LITERATURE SURVEY

TITLE: A Blockchain-Based Secure File Sharing System

YEAR OF PUBLISHING: 2019

AUTHOR NAME: X. Zhang, Y. Liu

ABSTRACT:

This paper proposes a secure file-sharing framework using blockchain technology to ensure data integrity and tamper-proof transaction logging. The system uses cryptographic hashing algorithms such as SHA-256 to generate digital fingerprints for files before storing them on a distributed ledger. The decentralized architecture improves transparency and eliminates reliance on third-party intermediaries. However, the system primarily focuses on file integrity and transaction security and does not incorporate multi-factor biometric authentication. Additionally, it lacks real-time anomaly detection or intelligent threat monitoring mechanisms.

TITLE: Multi-Factor Authentication System Using Facial Recognition and OTP

YEAR OF PUBLISHING: 2020

AUTHOR NAME: R. Kumar, S. Patel

ABSTRACT:

This research presents a multi-factor authentication (MFA) system integrating password verification, facial recognition using the Local Binary Pattern Histogram (LBPH) algorithm, and One-Time Password (OTP) validation. The proposed system enhances login security by combining knowledge-based, biometric, and possession-based authentication factors. Experimental results show improved resistance against brute-force and credential-stuffing attacks. However, the study focuses only on authentication mechanisms and does not address secure file storage, blockchain-based auditing, or behavioral threat detection.

TITLE: Digital Twin Technology for Cybersecurity Monitoring

YEAR OF PUBLISHING: 2021

AUTHOR NAME: L. Anderson, P. Roy

ABSTRACT:

This paper introduces the concept of a Digital Twin Network (DTN) for cybersecurity applications. The system creates a virtual replica of network infrastructure to monitor system behavior, detect anomalies, and predict cyber threats. Machine learning techniques are used to analyze login patterns, network traffic, and system events to identify suspicious activities. While the model demonstrates effective anomaly detection, it does not integrate blockchain-based logging or biometric authentication for secure access control.

TITLE: Secure Data Sharing Using Private Blockchain Architecture

YEAR OF PUBLISHING: 2022

AUTHOR NAME: H. Singh, T. Verma

ABSTRACT:

The study proposes a private blockchain framework for enterprise-level data sharing. The system ensures secure transaction recording, immutability, and controlled access within organizational environments. Smart contracts are implemented to automate access permissions and maintain transparent audit trails. Although the system improves data security and accountability, it does not incorporate multi-layer authentication mechanisms or real-time behavioral monitoring using Digital Twin concepts.

TITLE: Intelligent Threat Detection Using Behavioral Analysis

YEAR OF PUBLISHING: 2023

AUTHOR NAME: M. Brown, K. Wilson

ABSTRACT:

This research focuses on behavioral-based anomaly detection in secure systems. The proposed framework monitors login attempts, failed authentications, and abnormal access patterns to identify brute-force attacks and insider threats. Machine learning models are applied to detect deviations from normal user behavior. However, the system does not include blockchain technology for immutable logging nor biometric-based authentication, limiting its overall security robustness.

LANGUAGE DESCRIPTION

PYTHON:

Python is an interpreter, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding; make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed.

Often, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault. Instead, when the interpreter discovers an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace. A source level debugger allows inspection of local and global variables, evaluation of arbitrary expressions, setting breakpoints, stepping through the code a line at a time, and so on. The debugger is written in Python itself, testifying to Python's introspective power. On the other hand, often the quickest way to debug a program is to add a few print statements to the source: the fast edit-test-debug cycle makes this simple approach very effective.It ranges from simple automation tasks to gaming, web development, and even complex enterprise systems. These are the areas where this technology is still the king with no or little competence: Machine learning as it has a plethora of libraries implementing machine learning algorithms.Python is a one-stop shop and relatively easy to learn, thus quite popular now. What other reasons exist for such universal popularity of this programming language and what companies have leveraged its opportunities to the max?

Let’s talk about that. Python technology is quite popular among programmers, but the practice shows that business owners are also Python development believers and for good reason. Software developers love it for its straightforward syntax and reputation as one of the easiest programming languages to learn. Business owners or CTOs appreciate the fact that there’s a framework for pretty much anything – from web apps to machine learning. Moreover, it is not just a language but more a technology platform that has come together through a gigantic collaboration from thousands of individual professional developers forming a huge and peculiar community of aficionados. So what is python used for and what are the tangible benefits the language brings to those who decided to use it? Below we’re going to discover that. Productivity and Speed It is a widespread theory within development circles that developing Python applications is approximately up to 10 times faster than developing the same application in Java or C/C++. The impressive benefit in terms of time saving can be explained by the clean object-oriented design, enhanced process control capabilities, and strong integration and text processing capacities. Moreover, its own unit testing framework contributes substantially to its speed and productivity.

PYCHARM

PyCharm is a dedicated Python Integrated Development Environment (IDE) providing a wide range of essential tools for Python developers, tightly integrated to create a convenient environment for productive Python, web, and data science development.

Choose the best PyCharm for you

PyCharm is available in three editions:

- Community (free and open-sourced): for smart and intelligent Python development, including code assistance, refactorings, visual debugging, and version control integration.

- Professional (paid) : for professional Python, web, and data science development, including code assistance, refactorings, visual debugging, version control integration, remote configurations, deployment, support for popular web frameworks, such as Django and Flask, database support, scientific tools (including Jupyter notebook support), big data tools.

- Edu (free and open-sourced): for learning programming languages and related technologies with integrated educational tools.

- For details, see the editions comparison matrix.

Supported languages

To start developing in Python with PyCharm you need to download and install Python from python.org depending on your platform.

PyCharm supports the following versions of Python:

Python 2: version 2.7

Python 3: from the version 3.6 up to the version 3.10

Besides, in the Professional edition, one can develop Django, Flask, and Pyramid applications. Also, it fully supports HTML (including HTML5), CSS, JavaScript, and XML: these languages are bundled in the IDE via plugins and are switched on for you by default. Support for the other languages and frameworks can also be added via plugins

(go to Settings | Plugins or PyCharm | Preferences | Plugins for macOS users, to find out more or set them up during the first IDE launch).

Jupyter Notebook:

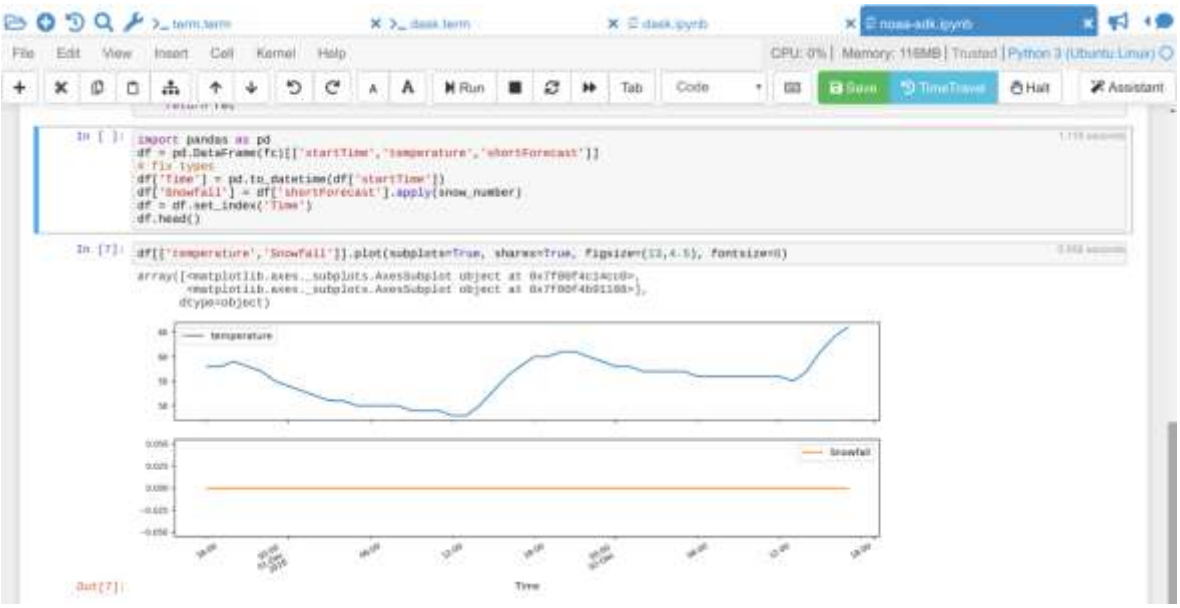
A Jupyter notebook is a specific filetype with the ending .ipynb, which records an interactive session with a Kernel. It made up of cells, which can either store one or more lines of code or formatted text. When you run a cell – which evaluates the piece of code in the cell via the active kernel session – you can see its output after the calculation is done. This combination of communicating back and forth with a kernel and adding descriptive text makes this form of document very attractive.

Jupyter Kernels

You can choose the programming language and environment by selecting a Jupyter kernel for the notebook. Popular choices are Python3, SageMath, and R. There many others. Our page on Jupyter Kernel Selection shows how to set the kernel.

JUPYTER NOTEBOOK BASIC:

By default, a Jupyter notebook on CoCalc has all CoCalc's core features, including real-time collaboration, side chat, and TimeTravel. Read more in our blogpost. The basic user interface looks like the following:



Above the main area is a menu bar and a button row:

- The menu bar contains all commands, and in particular the Kernel menu is for changing it if necessary.

- The button row gives you a one-click access to Run the current cell (otherwise press your Shift+Return keys), a way to restart the kernel (which clears the current session) and a Save button to make sure CoCalc has stored the file. The Time Travel button allows you to see previous versions of that notebook, such that you can go back in time to recover from a bad change.

- Active cell: in the screenshot above, the blue bar on the left and a blue border around a cell indicates that this is the currently active one. Actions like Run, Delete Cell, etc. operate on the currently selected cell. It is also possible to select more than one cell.

- Execution counter: On the left of each cell, there is an execution counter The number increases each time a cell is being run. After the kernel stopped and restarted, that counter starts again at 1.

- The output of code cells is below the input cell. For example, is the output of cell In the right hand corner of the input cell is some information about how long it took to calculate the result.

- Text cells are slightly different. Select “Markdown” in the dropdown menu in the button bar to change a code cell to such a markdown text cell. There, you can use Markdown to format the text. Similar to code-cells, either Run these text cells to see the processed Markdown code or press Shift+Return. To edit a text cell, either double click it or press your Return key.

- Saving: more general, the nice things about Jupyter Notebooks is that they save all your intput and output in one single file. This means you can download or publish the notebook as it is, and everyone else sees it in exactly the same way.

ANACONDA PYTHON

Anaconda® is a package manager, an environment manager, a Python/R data science distribution, and a collection of over 7,500+ open-source packages. Anaconda is free and easy to install, and it offers free community support.

Get the Anaconda Cheat Sheet and then download Anaconda.

Want to install conda and use conda to install just the packages you need? Get Miniconda.

Anaconda Navigator or conda?

After you install Anaconda or Miniconda, if you prefer a desktop graphical user interface (GUI) then use Navigator. If you prefer to use Anaconda prompt (or terminal on Linux or macOS), then use that and conda. You can also switch between them.

You can install, remove, or update any Anaconda package with a few clicks in Navigator, or with a single conda command in Anaconda Prompt (terminal on Linux or macOS).

- **To try Navigator**, after installing Anaconda, click the Navigator icon on your operating system's program menu, or in Anaconda prompt (or terminal on Linux or macOS), run the command anaconda-navigator.
- **To try conda**, after installing Anaconda or Miniconda, take the 20-minute conda test drive and download a conda cheat sheet.

Packages available in Anaconda

- Over 250 packages are automatically installed with Anaconda.
- Over 7,500 additional open-source packages (including R) can be individually installed from the Anaconda repository with the conda install command.
- Thousands of other packages are available from Anaconda.org.
- You can download other packages using the pip install command that is installed with Anaconda. Pip packages provide many of the features of conda packages and in some cases they can work together. However, the preference should be to install the conda package if it is available.
- You can also make your own custom packages using the conda build command, and you can share them with others by uploading them to Anaconda.org, PyPI, or other repositories.

Previous versions

Previous versions of Anaconda are available in the archive. For a list of packages included in each previous version, see Old package lists.

Anaconda2 includes Python 2.7 and Anaconda3 includes Python 3.7. However, it does not matter which one you download, because you can create new environments that include any version of Python packaged with conda. See Managing Python with conda.

CHAPTER 3. MODULE DESCRIPTION

The proposed Blockchain-Enabled Secure File Sharing Framework with Multi-Factor Biometric Authentication and Digital Twin-Based Threat Intelligence is divided into the following functional modules:

1. User Registration Module

This module allows new users to register into the system securely.

Functions:

- Collects user credentials (username, email, password).
- Captures and stores facial biometric data for LBPH-based recognition.
- Encrypts and securely stores user information in the database.
- Initializes blockchain identity for the user within the private network.

Purpose:

To create authenticated user accounts with biometric enrollment for enhanced security.

2. Multi-Factor Authentication Module

This module implements a three-level authentication mechanism to prevent unauthorized access.

Level 1: Password Verification

- Validates username and password.
- Protects against unauthorized login attempts.

Level 2: Facial Recognition (LBPH)

- Uses Local Binary Pattern Histogram (LBPH) algorithm.
- Matches live captured face with stored biometric template.
- Prevents identity spoofing.

Level 3: One-Time Password (OTP) Validation

- Generates time-based OTP.
- Sends OTP to registered email/mobile.
- Confirms possession-based authentication.

Purpose:

To provide strong layered security combining knowledge-based, biometric, and possession-based authentication.

3. File Upload and Integrity Verification Module

This module manages secure file storage and integrity protection.

Functions:

- Allows authenticated users to upload files.
- Generates SHA-256 hash value for each uploaded file.
- Creates a unique digital fingerprint of the file.
- Prevents file tampering and modification.

Purpose:

To ensure file authenticity and integrity through cryptographic hashing.

4. Private Blockchain Logging Module

This module ensures tamper-proof record maintenance.

Functions:

- Records authentication events.
- Stores file upload/download transactions.
- Saves generated SHA-256 hash values.
- Maintains immutable blocks linked through cryptographic hashes.

Purpose:

To provide transparency, traceability, and auditability of all system activities.

5. Digital Twin Network (DTN) Module

This module acts as a virtual behavioral replica of the system for proactive cybersecurity monitoring.

Functions:

- Monitors login attempts and failures.
- Tracks biometric mismatches.
- Observes OTP validation errors.

Purpose:

To enable predictive threat detection

6. Secure File Download Module

This module allows authorized users to securely access shared files.

Functions:

- Verifies user authentication status.
- Checks file integrity by re-validating SHA-256 hash.
- Logs download activity in blockchain.

Purpose:

To ensure secure and verified file retrieval without tampering.

CHAPTER 4. PROPOSED SYSTEM ANALYSIS AND DESIGN

1. Proposed System Analysis

1.1 Overview

The proposed system is a Blockchain-Enabled Secure File Sharing Framework integrated with Multi-Factor Biometric Authentication and a Digital Twin Network (DTN)-based threat intelligence layer. The system is designed to address the limitations of traditional password-based authentication by incorporating layered security, immutable audit logging, cryptographic integrity verification, and real-time anomaly detection.

The system ensures:

- Strong user authentication
- Secure file storage and transfer
- Tamper-proof transaction records
- Continuous behavioral monitoring
- Predictive threat mitigation

1.2 Problem Analysis

Traditional file-sharing systems suffer from:

- Single-factor authentication vulnerabilities
- Susceptibility to brute-force and phishing attacks
- Lack of tamper-proof audit trails
- Absence of real-time anomaly detection

- Limited transparency in access control

To overcome these issues, the proposed system integrates:

- Private blockchain architecture
- Multi-factor biometric authentication
- SHA-256 cryptographic hashing
- Digital Twin-based monitoring

1.3 Functional Requirements

1. User registration with biometric enrollment
2. Three-level authentication (Password + LBPH + OTP)
3. Secure file upload and download
4. SHA-256 hash generation for file integrity
5. Blockchain-based logging of all transactions
6. Digital Twin-based anomaly detection
7. Admin monitoring and alert system

1.4 Non-Functional Requirements

- High security and reliability
- Scalability for enterprise environments
- Low authentication latency
- Real-time threat detection capability

2. Proposed System Design

2.1 System Architecture Design

The system follows a layered architecture consisting of:

1. Presentation Layer

- User interface (Web-based dashboard)
- Login, Registration, Upload/Download pages

2. Authentication Layer

- Password verification
- LBPH facial recognition
- OTP validation

3. Application Layer

- File processing
- SHA-256 hash generation
- Blockchain interaction

4. Blockchain Layer

- Private blockchain network
- Block creation for each event
- Immutable transaction storage

5. Database Layer

- User credentials (encrypted)
- Biometric templates
- OTP logs
- File metadata

2.2 Authentication Design

The system implements three-factor authentication:

Step 1: Password Verification

- User enters username and password.
- System verifies encrypted credentials from database.

Step 2: LBPH Facial Recognition

- Captures real-time facial image.
- Compares with stored biometric template.
- Grants access if match threshold is satisfied.

Step 3: OTP Validation

- Generates time-based OTP.
- Sends to registered email.
- Verifies OTP before final login approval.

This layered approach reduces unauthorized access probability significantly.

2.3 File Integrity Design

- Upon file upload, system generates a SHA-256 hash.
- The hash acts as a unique digital fingerprint.
- Hash value is stored in the blockchain block.
- During download, hash is recalculated and compared.
- If mismatch occurs, file tampering is detected.

2.4 Blockchain Design

The system uses a Private Blockchain Network with the following structure:

Each block contains:

- Block ID
- Timestamp
- User ID
- Event Type (Login/Upload/Download)
- File Hash (if applicable)
- Previous Block Hash
- Current Block Hash

Features:

- Immutable linked structure
- Tamper-proof logging
- Transparent auditing

- Distributed verification

2.5 Digital Twin Network (DTN) Design

The DTN acts as a virtual behavioral model.

Monitored Parameters:

- Number of failed login attempts
- Biometric mismatches
- OTP validation failures

2.6 Data Flow Design (Logical Flow)

1. User registers → Biometric data stored
2. User login request → Multi-factor authentication
3. Successful login → Access dashboard
4. File upload → SHA-256 hash generation
5. Transaction stored in blockchain

2.7 Security Design

Security is implemented at multiple levels:

- Encryption of passwords
- Biometric template protection
- Cryptographic hashing (SHA-256)
- Blockchain immutability

- Role-based access control

3. Advantages of Proposed Design

- Eliminates single point of failure
- Prevents brute-force attacks
- Detects abnormal user behavior early
- Ensures complete auditability
- Enhances enterprise-level data protection
- Supports predictive cybersecurity

4. Conclusion of System Design

The proposed system integrates blockchain technology, biometric authentication, cryptographic integrity verification, and Digital Twin-based threat intelligence into a unified secure file-sharing architecture. The layered design ensures confidentiality, integrity, availability, and proactive security enforcement.

This design makes the framework suitable for enterprise data security, government applications, financial institutions, healthcare systems, and critical infrastructure environments.

CHAPTER 5. SYSTEM TESTING AND MAINTENANCE

1. System Testing

System testing ensures that the proposed Blockchain-Enabled Secure File Sharing Framework with Multi-Factor Biometric Authentication and Digital Twin Network (DTN) operates correctly, securely, and efficiently under various conditions. The testing process validates functional accuracy, security strength, performance stability, and threat detection capability.

1.1 Testing Objectives

- Verify correct implementation of three-level authentication.
- Validate SHA-256 file integrity verification.
- Ensure blockchain immutability and proper transaction logging.
- Test Digital Twin-based detection.
- Ensure reliability in enterprise-level environments.

1.2 Types of Testing

1. Unit Testing

Each module is tested independently to verify functionality.

Modules Tested:

- User Registration Module

- Password Authentication Module
- LBPH Facial Recognition Module
- OTP Generation & Validation Module
- File Upload & SHA-256 Hashing Module
- Blockchain Logging Module
- Digital Twin Monitoring

Purpose:

To ensure individual components work as expected.

2. Integration Testing

After unit testing, modules are integrated and tested together.

Test Scenarios:

- Login process (Password → Face Recognition → OTP)
- File upload → Hash generation → Blockchain storage

Purpose:

To verify seamless interaction between modules.

3. System Testing

The complete system is tested in a real-time simulated environment.

Test Cases Include:

- Valid user login and file access
- Invalid password attempts
- Facial mismatch attempts
- Incorrect OTP validation
- File tampering detection
- Abnormal download frequency detection
- Expected Outcome:
- Unauthorized users are denied access.
- Tampered files are detected.

4. Security Testing

Ensures robustness against cyber threats.

Security Tests:

- Brute-force attack simulation
- Replay attack testing
- Unauthorized file modification
- Blockchain tampering attempt
- SQL injection and input validation tests

Purpose:

To validate confidentiality, integrity, and availability of the system.

5. Performance Testing

Evaluates system performance under heavy load.

Parameters Tested:

- Authentication response time
- Blockchain transaction processing speed
- File upload/download latency

Goal:

Ensure system scalability for enterprise-level usage.

6. User Acceptance Testing (UAT)

End-users and administrators test the system to verify usability and functionality.

Focus Areas:

- Ease of login process
- Dashboard accessibility
- Alert clarity

2. System Maintenance

System maintenance ensures continuous performance, security updates, and adaptability to evolving cyber threats.

2.1 Types of Maintenance

1. Corrective Maintenance

- Fixing bugs or software errors.
- Resolving biometric recognition mismatches.
- Correcting blockchain transaction issues.

2. Adaptive Maintenance

- Updating authentication algorithms.
- Integrating new biometric technologies.
- Adapting to new security regulations.

3. Perfective Maintenance

- Improving system performance.
- Enhancing DTN anomaly detection accuracy.
- Optimizing blockchain storage efficiency.

4. Preventive Maintenance

- Regular security patch updates.
- Database backup and recovery testing.
- Blockchain integrity verification checks.
- Monitoring suspicious activity logs.

2.2 Backup and Recovery Plan

- Periodic database backups.
- Blockchain ledger replication.
- Disaster recovery mechanisms.
- Data restoration testing.

2.3 Security Maintenance

- Regular password policy updates.
- Biometric dataset re-training.
- Monitoring threat intelligence updates.
- Updating anomaly detection thresholds.

2.4 Future Enhancements

- AI-based predictive analytics integration.
- Advanced behavioral biometrics.

- Cloud-based blockchain deployment.
- Zero-Trust architecture implementation.

CHAPTER 6. METHODOLOGY

The proposed Blockchain-Enabled Secure File Sharing Framework with Multi-Factor Biometric Authentication and Digital Twin Network-Based Threat Intelligence is developed using a structured and layered methodology to ensure secure access control, data integrity, tamper-proof auditing, and proactive threat detection. The methodology begins with requirement analysis, where limitations of traditional single-factor authentication systems are identified, and security objectives such as confidentiality, integrity, availability, and transparency are defined. Based on these requirements, a modular system architecture is designed, integrating authentication, cryptographic processing, blockchain logging, and behavioral monitoring components.The implementation phase starts with user registration and biometric enrollment, where users provide credentials and facial data for future authentication. The system then implements a three-level authentication mechanism. In the first stage, password verification is performed using encrypted credential validation. In the second stage, facial recognition is carried out using the Local Binary Pattern Histogram (LBPH) algorithm to compare live facial input with stored biometric templates. In the third stage, a time-based One-Time Password (OTP) is generated and verified to confirm possession-based authentication. Only when all three levels are successfully validated is access granted to the user.Once authenticated, users can upload files securely. During the upload process, the system generates a SHA-256 cryptographic hash value that acts as a unique digital fingerprint of the file. This hash ensures data integrity by detecting any modification or tampering. All authentication events, file uploads, downloads, and hash records are then stored in a private blockchain network. Each block contains transaction details, timestamps, and cryptographic links to previous blocks, ensuring immutability and transparent auditability.

To enhance proactive cybersecurity, a Digital Twin Network (DTN) is integrated as a virtual behavioral replica of the system. The DTN continuously monitors user behavior, including failed login attempts, biometric mismatches, OTP validation errors. When abnormal behavior is detected, the system triggers alerts, temporarily restricts access if necessary.Finally, the system undergoes comprehensive testing, including functional testing, integration testing, security testing, and performance evaluation to ensure reliability and scalability. Through this structured methodology, the proposed framework achieves layered security, cryptographic integrity verification, immutable logging, and intelligent threat detection, making it suitable for enterprise-level and critical infrastructure applications.

CHAPTER 7. CONCLUSION

The proposed Blockchain-Enabled Secure File Sharing Framework integrated with Multi-Factor Biometric Authentication and Digital Twin Network-Based Threat Intelligence provides a comprehensive solution to address the growing challenges of cybersecurity in digital file-sharing environments. Traditional single-factor authentication systems are no longer sufficient to protect sensitive enterprise data from sophisticated attacks such as credential theft, brute-force attempts, insider threats, and unauthorized access. This framework overcomes those limitations by implementing a three-level authentication mechanism consisting of password verification, LBPH-based facial recognition, and OTP validation, thereby significantly strengthening access control.The integration of SHA-256 cryptographic hashing ensures file integrity by generating a unique digital fingerprint for every uploaded file. This mechanism effectively detects any unauthorized modification or tampering. Furthermore, the use of a private blockchain network guarantees immutable storage of authentication events, file transactions, and hash records, providing transparent, tamper-proof auditability and enhancing trust within the system.A key innovation of the proposed framework is the incorporation of a Digital Twin Network (DTN) for real-time behavioral monitoring. By continuously analyzing login failures, biometric mismatches, OTP errors, the system can proactively identify suspicious activities and prevent potential security. This predictive threat intelligence capability enhances adaptive security enforcement and strengthens overall system resilience.In conclusion, the combination of biometric authentication, cryptographic integrity verification, blockchain immutability, and Digital Twin-based anomaly detection creates a robust, intelligent, and scalable secure file-sharing environment. The proposed system not only ensures confidentiality, integrity, and transparency but also supports proactive cybersecurity measures, making it highly suitable for enterprise-level data protection and critical infrastructure security applications.

CHAPTER 8. FUTURE ENHANCEMENT

The proposed Blockchain-Enabled Secure File Sharing Framework with Multi-Factor Biometric Authentication and Digital Twin Network (DTN)-based Threat Intelligence provides a strong foundation for secure data management. However, several enhancements can be incorporated in the future to further improve scalability, intelligence, automation, and enterprise adaptability.

One major enhancement is the integration of Artificial Intelligence and Machine Learning-based anomaly detection. Instead of relying only on threshold-based monitoring, advanced deep learning models can be trained to identify complex behavioral patterns and zero-day attacks. This would significantly improve the predictive capabilities of the Digital Twin Network and reduce false positives.

Another future improvement is the adoption of behavioral biometrics, such as keystroke dynamics, mouse movement patterns, and device fingerprinting. Adding these continuous authentication mechanisms can strengthen security even after login, ensuring ongoing identity verification throughout the user session.The system can also be enhanced by implementing smart contracts within the private blockchain network. Smart contracts can automate access control policies, file-sharing permissions, expiration rules, and compliance auditing without manual intervention. This would increase efficiency and reduce administrative overhead.

To improve scalability and enterprise deployment, the framework can be migrated to a cloud-based or hybrid blockchain infrastructure. Distributed cloud nodes would enhance

performance, support high transaction throughput, and provide better disaster recovery capabilities. Additionally, containerization and microservices architecture can improve system flexibility and maintainability.Future versions of the system may incorporate Zero-Trust Architecture (ZTA) principles. Under this model, every access request is continuously verified regardless of network location. Combining zero-trust policies with multi-factor authentication and Digital Twin monitoring would create an even more resilient security ecosystem.

Another enhancement involves integrating advanced encryption mechanisms, such as homomorphic encryption or attribute-based encryption, to secure sensitive files while enabling controlled access. This would provide stronger confidentiality, especially for critical infrastructure and government applications.The Digital Twin Network can also be upgraded with real-time threat intelligence feeds from external cybersecurity databases. This would allow the system to adapt dynamically to emerging global threats and automatically update its anomaly detection models.

Finally, future work may include the development of a mobile-based secure access platform, enabling secure authentication and file access through smartphones using biometric sensors such as fingerprint or facial recognition. This would enhance usability while maintaining strong security standards.

In conclusion, by incorporating AI-driven analytics, behavioral biometrics, smart contracts, zero-trust security, advanced encryption, and cloud scalability, the proposed framework can evolve into a highly intelligent, adaptive, and enterprise-ready cybersecurity solution capable of addressing future digital threats.

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